Competency Questions (CQs)

This is a list of the Competency Questions (CQs) used to structure and validate the ClimOO Ontology suite, organized according to the scoping themes:

Types and characteristics of plastics, plastic debris, and microplastics

- What are the types of plastics and plastic debris? And related components?
- What are the characteristics of plastics and plastic debris? And microplastics?
- What are the polymers that compose each type of plastic?
- What distinguishes a plastic object from plastic debris?

Origins of plastic debris particles - Products of the corruption of plastic items; where plastic debris goes

- How is plastic generated/produced?
- What is the life cycle of plastic objects?
- What is plastic pollution?
- What are the processes leading/contributing to the formation of microplastics and other debris, especially in agriculture?
- What is a source of plastic debris? What are the different types of sources, especially in agriculture?
- What is a sink of plastic debris? What are the different types of sinks, especially in agriculture?

Climate, climate change, and related concerns, especially in agriculture

- What is a climate system? And its components?
- What is climate? And weather? And climate variability of a system of interest?
- What are the main processes involved in a climate system?
- What types of GHG emissions exist?
- What is climate change?

Extent of plastics and microplastics in agriculture today

- What are the uses of plastic in agriculture leading to the production of plastic debris, especially microplastics?
- Which agricultural activities require the use of plastic objects?

Threats of plastic pollution in agriculture related to climate - Possible alternatives to plastics

- How does climate change contribute to plastic pollution?
- How does plastic pollution contribute to climate change?

- What are the processes that can contribute to reducing plastic pollution, especially in agriculture?
- What of those processes do not exacerbate climate change?
- What of those processes also contribute to mitigating/reducing climate change?
- What are the alternatives to conventional plastics, especially in agriculture?